AMIN MEHRVARZ

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EDUCATION

Ph.D. Mechanical Engineering - Robotics/Mechatronics ***Sep 2017 - Aug 2021***
Northeastern University, Boston, MA, U.S **Overall GPA**: 3.87/4
Dissertation: Theoretical and Experimental Analyses of Tunable Active Metamaterials via Use of Feed-forward/back Controllers

M.Sc. Mechanical Engineering - Control ***Sep 2013 - Sep 2015***
Sharif University of Technology, Tehran, Iran **Overall GPA**: 3.82/4
Thesis: Positioning, tracking and vibrating boundary control in micro-beams with considering piezo-electric actuators effects.

B.Sc. Mechanical Engineering - Design and Manufacturing ***Sep 2008 - Aug 2013***
Sharif University of Technology, Tehran, Iran **Overall GPA**: 3.8/4
Thesis: Design and manufacture of a mold placement mechanism for a hydraulic press.

Diploma, Mathematics and Physics ***Sep 2004 - Jun 2008***
Dehkhoda, Karaj, Alborz, Iran **Overall GPA**: 4/4

RESEARCH INTERESTS

Control Theory,	Robotics,	Mechatronics,	Machine Learning
Multibody Dynamics,	Wave Propagation,	Vibrations	Acoustics

SELECTED COURSES

Robotics,	Advanced Control,	Advanced Vibrations
Mechatronics,	Linear Control,	Modal Analysis
Digital Signal Processing,	Nonlinear Control,	Vibrations of continues systems
Advanced Dynamics,	Adaptive Control,	Advanced Mechanisms Design
Machine Learning,	Optimal Control,	Advanced Mathematics

COMPUTER SKILL

MATLAB,	Python,	C++,	Simulink,	Simscape,	ROS2,
COMSOL,	Maple,	ANSYS,	SolidWorks,	CATIA,	AutoCAD

EXPERIENCE

Medtronic, Boston, MA, U.S. **Jan 2022–Present**
Sr. Software Engineer–Developer, Surgical Robotics Sep 2024–Present

- Project Lead: next-gen fully-wristed Tri-Staple integration into the Hugo software platform.
- Delivered the first software prototype with Unity simulation on the Hugo surgeon console for surgeon evaluation.
- Developed the control and kinematic software for the Tri-Staple surgical robot stapler instrument.

- train-Gauge Drop-Out Estimator: inferred torque & velocity from real sensor data at 98% accuracy for Tri-Staple.
- Validated the drop-out estimator via Monte Carlo sampling across learning-time and drop-out parameters.
- Real-Time Inverse Kinematics via Jacobian-Based Optimization: eliminated singularities for smoother haptic feedback.
- Developed low-jerk trajectory generators and closedloop motion controllers in MATLAB/Simulink for robotic surgical stapler instruments, enabling smooth position, velocity, and acceleration tracking.
- Designed and integrated real-time embedded control software for precision-driven robotic manipulators.
- Conducted hardware-in-loop testing, ensuring seamless integration between control software and medical devices.
- Collaborated with cross-functional teams (systems engineering, software, and validation) to define and review safety-critical software requirements.
- Reviewed MATLAB/Simulink models, verifying compliance with precise standards for medical device software quality.
- Participated in design reviews and contributed to the documentation of architecture, unit testing, and anomaly tracking.
- Integrated mechanical, electrical, and software components, particularly in an embedded, high-performance control environment.

Sr. Software Engineer–Test Lead, Surgical Robotics Jun 2023–Sep 2024

- Test Lead for software release in surgical robot and leading software verification for FDA submission
- Provide technical leadership for the software verification engineering verifying the functionality of feedback control loop algorithms designed and developed in MATLAB/Simulink
- Lead a team verifying controls software for robotic manipulators and surgical instruments
- Lead a team integrating and testing new controller designs on hardware and with the rest of the robotic platform
- Conduct software and product hardware-in-loop testing and create model and prototypes
- Review teammates MATLAB/Simulink models and ensure that all software meets quality standards; prepare and participate in design reviews
- Document or participate in the documentation of the software including verification test plans and reports
- Lead the definition and implementation of different software testing techniques to increase test coverage in both automation and manual testing.
- Understand the technical behavior of the system, architecture, software requirements with an aptitude to identify product defects in all phases of development.
- Lead development of solutions that have significant improvements of processes and test development with aptitude for continue improvement.

- Work with the Project Management Office to coordinate dry runs and formal verification activities as required
- Collaborate with technical engineering teams, mentor junior test engineers.

Controls Verification Engineer, Surgical Robotics Jan 2022–Jun 2023

- Conducted software and producted hardware-in-loop testing and created model and prototypes
- Reviewed MATLAB/Simulink models to ensure that all software meets quality standards
- Documented and participated in the documentation of the software including verification test plans and reports
- Implemented different software testing techniques in Python and Matlab to increase test coverage in both automation and manual testing.
- Analyzed the technical behavior of the system, architecture, software requirements with an aptitude to identify product defects in all phases of development.
- Led development of solutions that have significant improvements of processes and development with aptitude for continue improvement.
- Worked with the Project Management Office to coordinate dry runs and formal verification activities as required
- Collaborated with technical engineering teams, mentor junior test engineers.
- Developed the verification test methods to evaluate performance metrics of complex, electromechanical medical systems

Northeastern University, Boston, MA, U.S **Sep 2017–Dec 2021**

Graduate Research Assistant *Sep 2017–Dec 2021*

- Conducted doctoral research in robotics, control systems, and mechatronics, with an emphasis on dynamic modeling, feedback control, and tunable active metamaterials.
- Developed and analyzed mathematical models and control strategies using MATLAB and Simulink to investigate the dynamic behavior and performance of complex mechanical systems.
- Applied theoretical analysis, numerical simulation, and experimental validation to evaluate control methods and system performance.
- Disseminated research findings through peer-reviewed publications and conference presentations, contributing to 10 journal articles and 7 conference papers.
- Served as an instructor for the following courses: Advanced Dynamics, Advanced vibration, Mechatronics, and Measurement Laboratory.
- Coordinated the activities and operations of the Piezoactive Systems Laboratory (PSL).

Sharif University of Technology, Tehran, Iran **Sep 2013 – Sep 2015**

Graduate Research Assistant *Sep 2013 – Sep 2015*

- Conducted research in nonlinear control systems, partial differential equation (PDE)-based control, adaptive control, and fluidstructure interaction.
- Applied mathematical modeling, numerical simulation, and computational methods to investigate dynamic systems, vibration suppression, and control system stability.
- Published 3 journal articles and 1 conference paper based on research findings.

- Delivered instruction in undergraduate engineering courses, including Automatic Control, Dynamics of Machinery, Vibrations, Dynamics and Vibrations Laboratory, and Mechanism Design.

Power Planet Repairs, Internship, Tehran, Iran **Jun 2013 – Aug 2013**

- Modeled and redesigned an 8000kg gas turbine with Catia and rebuilt the model in Solidworks
- Simulated the model of the gas turbine by SolidWorks and ANSYS with different loads
- Analyzed the damages on the system and determined the effects of the cracks on the system's failure
- Presented a SolidWorks workshop to improve the knowledge of the company's staff

Sharif University of Technology, Internship, Tehran, Iran **Jun 2011 – Aug 2011**

- Research on different available FEA software including ANSYS, Fluent, SolidWorks's modules & Comsol Multiphysics
- Modified an available solenoid valve and documented the steps of it
- Calculated and Analyzed the solenoid valve flow rate using SolidWorksFlow simulation
- Compared the results of the modified valve with the original one, and showed performance improvement by

PUBLICATIONS

Google Scholar

- Mehrvarz, A., Khodaei, M. J., Darabi, A., Zareei, A., & Jalili, N. (2024). Experimental demonstration of broadband reconfigurable mechanical nonreciprocity. Journal of Vibration and Control, 30(1-2), 288-294.
- Khodaei, M. J., Mehrvarz, A., Ghaffarivardavagh, R., & Jalili, N. (2022). Acoustic meta-silencer: Toward enabling ultrawide-band air-permeable noise barrier. Journal of Vibration and Control, 28(15-16), 1928-1936.
- Khodaei, M. J., Mehrvarz, A., Ghaffarivardavagh, R., & Jalili, N. (2022). Retrieving effective acoustic impedance and refractive index for size mismatch samples. AIP Advances, 12(6).
- Korivand, S., Mehrvarz, A., Candelino, N., Khodaei, M. J., & Jalili, N. (2021). Band Gap and Stiffness Tuning of a Sandwich Plate by Magnetostrictive Materials. ASME Letters in Dynamic Systems and Control, 1(4), 041007.
- Mehrvarz, A., Khodaei, M. J., Clark, W., & Jalili, N. (2020, November). A new dynamic model of a two-wheeled two-flexible-beam inverted pendulum robot. In ASME International Mechanical Engineering Congress and Exposition (Vol. 84553, p. V07BT07A044). American Society of Mechanical Engineers.
- Khodaei, M. J., Mehrvarz, A., Ghaffarivardavagh, R., & Jalili, N. (2020). Non-centric cavity-based acoustic metasurface: Enabling full phase modulation. AIP Advances, 10(9), 095020.
- Khodaei, M. J., Candelino, N., Mehrvarz, A., & Jalili, N. (2020). Physiological closed-loop control (PCLC) systems: Review of a modern frontier in automation. Ieee Access, 8, 23965-24005.
- Salehzadeh, R., Candelino, N., Khodaei, M. J., Mehrvarz, A., & Jalili, N. (2020, October). Parameter Sensitivity Analysis of Piezoelectrically-Actuated Flexural/Torsional Vibrating Beams. In Dynamic Systems and Control Conference (Vol. 84287, p. V002T38A008). American Society of Mechanical Engineers.

- Khodaei, M. J., Inaloo, M. H. B., Mehrvarz, A., & Jalili, N. (2019). An Adaptive Neuro-fuzzy Strategy in Closed-loop Control of Anesthesia. arXiv preprint arXiv:1910.06325.
- Ardekany, A. N., Daneshmand, F., & Mehrvarz, A. (2019). Vibration Analysis of a Micropolar Membrane in Contact with Fluid. Iranian Journal of Science and Technology, Transactions of Mechanical Engineering, 43(1), 695-704.
- Mehrvarz, A., Najafi Ardekani, A., Khodaei, M. J., & Jalili, N. (2019). Vibration analysis and control of fluid containers using piezoelectrically-excited side wall. Journal of Vibration and Control, 25(7), 1393-1408.
- Khodaei, M. J., Mehrvarz, A., & Jalili, N. (2019). An adaptive multi-critic neuro-fuzzy control framework for intravenous anesthesia administration. IFAC-PapersOnLine, 51(34), 202-207.
- Mehrvarz, A., Salarieh, H., Alasty, A., & Vatankhah, R. (2018). Boundary vibration control of strain gradient timoshenko micro-cantilevers using piezoelectric actuators. arXiv preprint arXiv:1812.01155.
- Khodaei, M. J., Mehrvarz, A., Candelino, N., & Jalili, N. (2018, September). Theoretical and experimental analysis of coupled flexural-torsional vibrations of rotating beams. In Dynamic Systems and Control Conference (Vol. 51913, p. V003T42A004). American Society of Mechanical Engineers.
- Mehrvarz, A., Khodaei, M. J., Clark, W., & Jalili, N. (2018, September). Modeling and dynamics analysis of a beam-hoverboard self-transportation system. In Dynamic Systems and Control Conference (Vol. 51913, p. V003T32A008). American Society of Mechanical Engineers.
- Mehrvarz, A., Salarieh, H., Alasty, A., & Vatankhah, R. (2018). Vibration boundary control of Timoshenko micro-cantilever using piezoelectric actuators. Scientia Iranica, 25(2), 711-720.
- Ardekany, A. N., & Mehrvarz, A. (2016, October). Vibrations suppression of a Euler-Bernoulli beam in contact with a fluid using piezoelectric actuators. In 2016 4th International Conference on Robotics and Mechatronics (ICROM) (pp. 285-288). IEEE.

PROJECTS

Control of Active Metamaterial

- Proposed and designed a feedback/forward controller to spatiotemporal modulate the metamaterials stiffness using MATLAB
- onfigured and analyzed wave propagation in the controlled system with %15 variable stiffness using MATLAB/COMSOL
- Studied and validated wave propagation experimentally using piezo sensors/actuators, hardware-in-the-loop, Simulink & dSPACE
- Analyzed the real-time data and investigated the nonreciprocity feature of the controlled system using MATLAB/Simulink

Closed-Loop Control of Anesthesia

- Proposed and advanced a medical device controller with %20 drug consumption reduction in anesthesia using Python
- Optimized and trained the adaptive multi-critic neuro-fuzzy controllers coefficients using Python
- Designed an optimal and adaptive controller for automation in anesthesia with no overshoot using Python

Moving Robot

- Analyzed and compared PID and MPC controllers effects on the stability of a flexible robot using MATLAB
- Developed and designed arm robot gripper to detect different types of fish using machine learning/Python

Control of active baffles

- Innovated and introduced a smart fluid tank to eliminate the %99 of fluid movements within 0.01s using FEA/MATLAB
- Presented a robust nonlinear controller to stabilize the fluid movement using feedback linearization techniques
- Compared the designed feedback controller with the available products and showed the %30 more efficiencies of it using MATLAB

Acoustics wave manipulation

- Developed and initiated a metasurface for complete phase modulation using MATLAB/COMSOL
- Presented and investigated a methodology for broad-band sound attenuators in air-permeable structures
- Developed a theory-simulation method to measure the acoustic properties of metamaterials using MATLAB/COMSOL

PDE controllers for Atomic Force Microscopy (AFM)

- Examined and compared the effects of different PDE controllers on AFM using C++
- Presented and simulated an ODE feedback controller to eliminate the vibrations of AFM in 0.001s using C++
- Introduced and implemented a Lyapunov-based boundary controller to %99 stabilize the movement of AFM using C++

Control of Coronary Artery Model

- Designed and evaluated chaos controller to reduce the error of coronary artery system to %1 using MATLAB/Simulink
- Developed a sliding mode model reference, adaptive controller, for the coronary artery's nonlinear model using Matlab/Simulink

Fluid-Structure Vibrations

- Investigated and analyzed the vibrations of micropolar membranes in contact with fluid using C++
- Monitored and interpreted rotor-shafts behavior under the influence of fluid forces using MATLAB

PROCEEDINGS CONFERENCES

- International Mechanical Engineering Congress and Exposition, ASME 2020, Virtual, USA
- Dynamic Systems and Control Conference, ASME 2020, Virtual, USA
- Learning for Dynamics and Control (L4DC), 2019, Cambridge, MA, USA
- 2nd IFAC Conference on Cyber-Physical and Human Systems CPHS 2018, Miami, Florida, USA
- Dynamic Systems and Control Conference, ASME 2018, Atlanta, Georgia, USA

- 4th International Conference on Robotics and Mechatronics (ICROM), IEEE 2016, Tehran, Iran

LANGUAGE

Persion (Native) English (Advanced) Arabic (Beginner)

HONNORS AND AWARDS

- Awarded Outstanding Graduate Researcher, College of Engineering, Northeastern University, 2020
- Awarded Outstanding Teaching, College of Engineering, Northeastern University, 2021
- Awarded Outstanding Reviewer, IOP Publishing, 2020
- Awarded scholarship, Mechanical and Industrial Engineering Dept, Northeastern University, 2017
- Ranked 14th out of 20,000 participants, Nationwide entrance exam for M.Sc. degree, 2013
- Ranked 196th out of 350,000 participants, Nationwide Entrance exam for B.Sc. degree, 2008

EXTRACURRICULAR ACTIVITIES

- American Society of Mechanical Engineers (ASME) Member
- International Federation of Automatic Control (IFAC) Member

REVIEWER EXPERIENCE

- Reviewer of Smart Materials and Structure Journal
- Reviewer of IEEE Access Journal
- Reviewer of Journal of Vibration and Control
- Reviewer of 2nd IFAC Conference on Cyber-Physical and Human-Systems 2018
- Reviewer of the 4th RSI International Conference on Robotics and Mechatronics (ICROM)